# The Box in Balance

Student sheet

**Name:** _____ **Date:** _____

**Safety:** I hold the box in both hands, over the table. The cable stays slack. I never throw the box to a classmate: I put it down on the table.

## 1. What I see on the screen

On the screen you can see three numbers: **AX**, **AY** and **AZ**. The box feels whatever holds it up **pushing it upwards** so that it does not fall — the table, or your hand. The numbers show how big that push is in each of the three directions.

The activity starts with the **Start** button. You move on with the green button at the bottom right, you go back with **Back**, and at the last step the green button reads **Finish**.




## 2. What I do on my own

**First I predict. I do not change my prediction after I measure.**

**1.** The box is lying still on the table. I think AZ will show:
☐ zero ☐ something other than zero if you can, write the number too: _____

because _____

_____

**2.** When I tilt the box to the right, I think AX will:
☐ go up ☐ go down ☐ stay the same

because _____

_____

**Now I measure.** I put the box down, wait for it to be still, count to three, then write the numbers exactly as they appear on the screen.

| Position of the box | AX | AY | AZ |
|---|---|---|---|
| Flat on the table | | | |
| Tilted to the right | | | |
| Tilted to the left | | | |
| Tipped forward | | | |
| On its edge | | | |
| Upside down | | | |
| Flat, a second time | | | |

When I tilt the box, the number AX: ☐ ↑ ☐ ↓ ☐ →
(tick one: goes up, goes down, stays the same)

**Challenge.** Can you make all three numbers zero at the same time? ☐ YES ☐ NO

## 3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

The difference is explained by _____

_____

**Write one sentence:** why the box does not show zero when it is sitting still.

_____

**Exit ticket**

Write one thing that surprised you today.

_____